

Hemaang Sood

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[Github](#): hemaangsood | [Portfolio](#): hemaangsood.github.io

Skills

AI & ML

PyTorch, TensorFlow, OpenCV, OpenAI Gym, OpenAI API, HuggingFace(Transformers), xgboost.

Data Analysis

Pandas, Numpy, Matplotlib, Plotly, Tableau, SQL.

Languages

Python, JavaScript, Go, Java, Kotlin, R.

Databases

MongoDB (Atlas), PostgreSQL (Supabase), Supabase, FAISS, Qdrant, Redis.

Cloud, DevOps & Tools

AWS (EC2, S3), Git, GitHub, Docker, Nginx, Postman.

CI/CD & Testing

GitLab CI/CD, Docker-in-Docker, pytest, Poetry, PostGIS.

Experience

Synechron

Jr Associate

- Completed introductory sessions on **Capital Markets** and **Commodities**, developing foundational understanding of financial systems and trading concepts.
- Built a full-stack project for an **Amusement Park Management System** using **MERN** with **Vite.js** and **Tailwind CSS**, focusing on modular UI design, API integration, and database connectivity.
- Built a backend server project for an **Insurance Management System** using Springboot, focusing scalability security and modularity.

Bharat Electronics Limited (BEL)

Intern

- Developed an Authentication and Authorization module for DRDO as part of the internship as part of a secure intranet website.
- Worked on a Large Language Model (LLM) interface to locally process and extract insights from sensitive documents using **RAG**. Integrated a **voice input pipeline** using Vosk to transcribe user speech into queries, enhancing accessibility and usability.

Education

University of Waterloo

Electrical & Computer Engineering, MEng

Jan 2026 – April 2027

GPA - 3.9 / 4.0

BMS Institute of Technology and Management (BMSIT&M)

Bachelor of Engineering in Computer Science and Engineering

2021 – 2025

CGPA - 8.9 / 10.0

Conference Publications & Presentations

- Hemaang Sood et al., "Improving Face Recognition for Smart Home Security: A Study of ResNet, Siamese Networks, and Attention Mechanisms" Presented at IEEE COMPSIF 2025.

Personal Projects

Virtual POCT — AI-Powered Quality Control Assistant:

Individual

- Built a full-stack AI-powered quality control platform for Point-of-Care Testing (POCT), enabling healthcare operators to troubleshoot devices through a conversational assistant with auditable QC workflows.
- Designed a streaming WebSocket architecture using FastAPI and LangChain to deliver real-time LLM responses while supporting concurrent multi-session conversations.

- Implemented a deterministic quality control evaluation engine that combines operator input with database context to classify QC scenarios and automate compliance actions.
- Developed a Dockerized microservice architecture with React, FastAPI, PostgreSQL, and Nginx, including JWT authentication, role-based access control, and persistent session history.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Docker, Nginx, WebSockets, LangChain, Google Gemini, JWT.

Facial and Voice Recognition-Based Door Lock System:

Group

- Achieved 98.7% accuracy in preventing unauthorized access by integrating facial and speech recognition algorithms for secure authentication.
- Processed and augmented the YaleFace dataset, utilizing a custom YOLO model to extract, align, and normalize facial features for improved model training.
- Implemented a Siamese Network to optimize feature comparison, enhancing the precision and robustness of facial recognition for reliable access control.
- Built Backend in Django and made a NGINX reverse proxy.
- Published paper in **IEEE Compsif 2025** for the same.
- **Technologies and Tools:** OpenCV, PyTorch, Google Speech API, RESTful APIs, Python, Raspberry Pi, Hybrid Neural Networks, Docker, React, PostgreSQL, Tailwind, Shad CN, AWS EC2, Django, NGINX.

Resume Builder:

Individual

- Built an AI-powered resume optimization tool that parses LaTeX resumes, tailors content to job descriptions, and generates polished, job-specific resume output.
- Agent utilizes multi-step process to parse optimize and output the the optimized resume.
- **Technologies Used:** Python, OpenAI API, LaTeX, SQLite, Jinja2, Rich, pdflatex, python-dotenv

Stock Analysis using Machine Learning:

Individual

- Collected, cleaned, and transformed historical stock data for 25 stocks, integrating 10 global economic factors to enhance model performance and generate actionable insights.
- **Technologies and Techniques:** LSTM, GRU, CNN, Reinforcement Learning, Hybrid Neural Networks, PyTorch, Python, AWS (EC2, S3), Pattern Recognition, Data Preprocessing, Feature Engineering, Statistical Analysis.

Reinforcement Learning for Snake Game:

Individual

- Created Convolutional Neural network that learnt to complete snake game within 45 minutes of training.
- **Technologies Used:** Python, TensorFlow, Deep Q-Learning, RL, OpenAI Gym, PyGame, CNN, NumPy, Matplotlib.

Routed: Travel Partner Recommendation System

Group

- Designed and developed a microservices-based platform to match travelers with compatible trip partners using trip overlap, preferences, and geospatial proximity.
- Implemented a FastAPI backend with JWT authentication, trip management APIs, and a matching system powered by PostgreSQL and PostGIS for spatial queries.
- Built an event-driven notification pipeline using Kafka and a Go-based worker service that processes match events and stores notifications in MongoDB.
- Built an Kotlin microservice with coroutines to extract semantic embeddings from bios.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, PostGIS, Kafka, MongoDB, Docker, Go, SQLAlchemy.